

RAJ STUDIO

The complete beginner's user guide

An AI music generation studio built on ACE-Step V1.5

Composition · Generation · Remix · Repaint

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Version 1.0 · October 2025

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Welcome to Raj Studio

Raj Studio is a modern, dark-themed web studio for the ACE-Step V1.5 open-source music generation model. It lets you turn a description, a set of lyrics, or an existing audio file into a full musical track — without any command line, Gradio quirks, or model plumbing to worry about.

This guide walks you through the interface, explains every meaningful control in plain English, and shows you the exact recipe for turning a set of lyrics into a polished, beat-tight MP3.

Who this guide is for: anyone who's just opened Raj Studio for the first time. No music-theory background needed. If you know what a chorus is and can copy-paste a prompt, you can follow this end-to-end.

Prerequisites

- The Raj Studio frontend is open at **`http://localhost:5173`**
- The ACE-Step backend is reachable at **`http://127.0.0.1:8001`**
- The header status pill in the top-right reads **Connected** in green

If the pill is red or says 'Offline', jump to **Section 7 · Troubleshooting** before continuing.

1. What you see on screen

When Raj Studio loads, the interface is organised top-to-bottom into six regions:

Region	Purpose
Top strip	Attribution — Developed by Rajeshwar Singh · Powered by ACE-Step V1.5
Header	Hamburger menu (mobile), backend status pill (green = connected, red = offline), Settings shortcut
Sidebar	Studio, Generate, Projects, History, Audio Library, Settings, About
Hero banner	Big RAJ STUDIO logo, live status chips, animated waveform — purely visual
Generation mode	Pick one of four modes (Simple / Custom / Remix / Repaint). Drives what fields appear.
Form area	Composition inputs — differs by mode
Right column	Live Progress panel while generating, Results with a player when done

The **green Connected pill** in the top-right is the single most important thing to watch. If it says Offline in red, the backend isn't reachable — fix that first.

2. The four generation modes

The mode selector at the top of the Studio page is the biggest single decision you'll make. Every other control adapts to the mode you pick.

Mode	Use it when	What it does
Simple	You have a vibe in words but no lyrics yet, and want the AI to draft everything	You type a plain-English description. The 5Hz LM drafts caption, lyrics, BPM, key and language automatically. Best for exploring.
Custom	You know what you want — you have lyrics or a strong prompt	Full manual control over prompt, lyrics, BPM, key, duration, guidance, etc. This is the mode you'll live in as you get better.
Remix	You have an existing audio file and want the same song in a different style	You upload a track, describe the new style ('turn this into synthwave'), model rebuilds it.
Repaint	You have an existing track and want to regenerate only part of it	Upload track, specify start/end seconds, model replaces only that slice. Great for fixing one bad verse.

Rule of thumb: Simple for exploring, Custom for producing, Remix and Repaint for editing existing work.

3. The fields, in plain English

The three creative fields

Field	Plain English	Example
Music caption	What the song sounds like — genre, instruments, mood, energy	<i><i>Warm indie folk with fingerpicked acoustic guitar, gentle brushed drums, soft female harmonies, nostalgic feel</i></i>
Lyrics	The actual words. Use <code><code>[verse]</code></code> , <code><code>[chorus]</code></code> , <code><code>[bridge]</code></code> tags to structure.	See Section 5 below
Negative prompt	What to avoid . Rarely needed.	<i><i>harsh distortion, robotic vocals, mumbling</i></i>

The structural fields (Optional parameters)

Field	What it does	Leave Auto if...
BPM	Beats per minute — the tempo	You don't have a specific tempo in mind. LM picks a value that fits the genre.
Key / scale	Musical key (e.g. C major, E $\blacksquare$ minor)	You don't read music.
Time signature	Rhythm feel (4/4 for most pop, 3/4 for waltz, 6/8 for ballads)	Almost always leave Auto unless you're doing something unusual
Vocal language	Language of the vocals	You want the model to detect it from your lyrics
Duration (s)	Song length in seconds	You want the model to pick a natural length. Otherwise, 60 to 120 seconds is a good short-song range.

Every field with a small purple **BPM AUTO** style badge underneath is currently in auto mode. Click **Reset all to Auto** at the top of that section to clear all your overrides at once.

The quality & control fields

Field	What it does	Beginner default
Batch size	How many takes to render in one job	2 — you'll want to compare a couple
Inference steps	Refinement passes the diffusion model runs. More = higher quality, slower	27 (turbo) or 60 (base model)
Guidance scale	How strictly the model follows your prompt. Higher = obeys prompt more, less creative	7
Audio format	Output file format	mp3 for sharing, wav for editing, flac for lossless archival
Seed	Random number that determines the exact generation	Leave Random seed on. Only pin a seed to reproduce a specific take.

The AI-assisted section (LM code hints)

These all require the 5Hz language model to be loaded on your backend (ACESTEP_INIT_LLM=true in your .env).

Toggle	What it does
Think (LM-DiT)	Uses the LM to generate audio codes first, then diffusion refines them. Higher quality, slower.
AutoGen	Fills in missing caption / lyrics / metadata for you before generation starts
Enhance input first	Rewrites your caption and lyrics through the LM to make them clearer for the model
Retake hint	Nudges the LM to produce a <i><i>variation</i></i> rather than a rebuild
Edit hint	Nudges the LM to <i><i>edit</i></i> rather than fully regenerate

The power-user section (Advanced generation parameters)

Ignore these until you've done ~10 generations. They control inference internals: ODE vs SDE, CFG intervals, DCW wavelets, tiled decoding. The defaults are safe.

4. Your first generation, in 90 seconds

Follow this exactly the first time you use Raj Studio. It's designed to fail-fast if anything is misconfigured.

- Make sure the header pill is **green Connected**.
- Leave mode on **Custom**.
- Click **Random sample**. The form fills with a working example.
- Set **Duration** to **10**. You want your first generation to be fast so you can see the whole pipeline work.
- Click the giant **Generate music** button.
- Watch the Progress panel on the right: *queued* → *running* → *complete*. First run takes 30 to 90 seconds on Apple Silicon because the model has to warm up.
- Play the result. Click **Download**. That's it — you've made an AI track.

Now do it again with your own prompt and lyrics, and read the workflow in Section 5.

5. The lyrics-first workflow

You have lyrics. You want a great backing track with tight beats. This is the exact recipe.

Step 1 — Format your lyrics correctly

The model responds massively better to structured lyrics. Use section tags on their own lines, and leave a blank line between sections.

```
[verse]
We were dancing in the corner of the room
Where the neon signs were bleeding into you
Every heartbeat sounded like a lullaby
Every promise felt like it could never die

[chorus]
Take me back to summer
Take me back to gold
Where the nights were younger
And we'd never grow old

[verse]
Now the streets are quiet and the lights are dim
And I trace your name across the window's rim

[chorus]
...

[bridge]
Maybe we were only borrowing time
Maybe love was just a rhyme

[outro]
Take me back... take me back...
```

Supported tags: [verse], [chorus], [bridge], [intro], [outro], [pre-chorus], [hook], [refrain]. A blank line between sections matters.

Paste this into the **Lyrics** field.

Step 2 — Write a specific Music caption

Vague prompts = vague songs. This is the single biggest lever you have.

Level	Example
Weak	<i>pop song</i>
Good	<i>upbeat synth-pop with punchy 808 kick, side-chained pluck synths, airy female lead vocals with reverb, catchy chorus lift</i>
Great	<i>mid-2010s synth-pop in the style of CHVRCHES: driving 4-on-the-floor kick at 118 BPM, side-chained pluck synths, warm analog bass, airy female lead vocals with plate reverb, bright pre-chorus build, wide layered chorus with vocal stacks and clap on 2-and-4</i>

The formula: genre → tempo hint → key instruments → vocal style → production feel.

Then click **Enhance caption**. The LM will refine your description and often add production terms that improve the output.

Step 3 — Tell the model about the beats

For "perfect beats," the four fields that matter most:

Field	Value	Why
BPM	Set it deliberately (see table below)	Auto sometimes picks a tempo that fights your lyric syllable density
Time signature	4/4 for almost everything modern; 6/8 for ballads and swing; 3/4 for waltz	Wrong signature = the beat won't lock to the vocals
Guidance scale	7 to 9	Higher = the model respects your <i>punchy kick</i>, <i>clap on 2-and-4</i> phrases from the caption
Inference steps	40+ (turbo) or 80+ (base)	More steps = tighter transients on drums, less smearing

BPM cheat sheet by genre

Genre	Range	Sweet spot
Hip-hop / R&B	70–100	85
Chill / lo-fi	70–90	80
Reggaeton / dembow	90–100	95
House / dance-pop	118–128	122
Synth-pop / indie-pop	110–125	118

Genre	Range	Sweet spot
Rock	120–140	128
Drum & bass	165–180	174
Ballad	60–75	68

Step 4 — Set duration to match your lyrics

Roughly count: **1 line** $\approx$ **2 to 3 seconds** at 120 BPM in 4/4.

- Short jingle: 15 to 30 seconds
- One verse + chorus loop: 60 seconds
- Verse–Chorus–Verse–Chorus: 90 to 120 seconds
- Full song (V-C-V-C-B-C): 180 to 240 seconds

Longer = slower to generate and more chances for the model to drift. If your song is over 90 seconds, use **batch size = 1** to save time and iterate faster.

Step 5 — Multiple takes, pick the best

Set **Batch size** to **2** or **3**. Cost is $\sim 2\text{--}3\times$ the time but you get variants and can pick the strongest. Set it back to 1 once you find a caption that reliably nails it.

Step 6 — Pick the right output format

Format	Use when
wav	You'll edit / master in a DAW (Logic, Ableton, FL). Zero quality loss.
flac	Archival. Same quality as wav, $\sim$ half the size.
mp3	Direct sharing (Discord, WhatsApp, mobile). 'Good enough' but lossy.
wav32	Overkill unless you're doing serious mastering. Use wav.
opus / aac	Only if a specific platform requires them.

Best-in-class MP3 workflow: generate as **wav** or **flac**, then encode to MP3 externally with `ffmpeg -i input.wav -c:a libmp3lame -b:a 320k output.mp3` if you want maximum MP3 quality. The direct **mp3** output uses a fixed bitrate that's slightly lower.

6. Best-in-class recommended preset

Copy these values directly for your first serious generation. This is the setting I recommend after hundreds of generations across genres.

Field	Value	Why
Mode	Custom	Full control
Music caption	Detailed, ~40 words, using the formula above	Model needs specificity
Lyrics	Tagged with [verse], [chorus], blank lines between	Structure matters
Instrumental	Off (unless you want no vocals)	—
BPM	From cheat sheet above	Don't leave Auto here
Key / scale	Auto	Let the model choose what fits vocals
Time signature	4/4	Right for 95% of modern music
Vocal language	Match your lyrics language	Improves pronunciation
Duration	90 for a full short song	Long enough for structure
Batch size	2	Compare takes
Inference steps	40 (turbo) / 80 (base)	Sweet spot: much better than default, still reasonable time
Guidance scale	8	Tight adherence to your caption
Audio format	wav then convert to MP3 externally	Best quality
Seed	Random, then pin the one you like	Reproducibility
Enhance caption	Click once before generating	Free quality bump
Random seed	On	For batch variety
Think mode	On (if LM loaded)	Better rhythm / melody structure
Tiled decode	On	Prevents OOM on Apple Silicon

Expected generation time on an M-series Mac (17.8 GB unified memory): ~2 to 4 minutes per 90-second track at these settings.

7. Troubleshooting common problems

If something goes wrong, check this table before anything else.

Problem	Cause	Fix
Vocals sound mumbled / language wrong	Model guessed the wrong language	Set Vocal language explicitly, don't leave Auto
Beat doesn't lock to lyric phrasing	BPM too fast or too slow for syllable density	Try BPM ± 10 from your first attempt
Song feels shapeless, no chorus lift	Lyrics not tagged	Add [verse] / [chorus] tags with blank lines
Guitar / piano sounds muddy or smeary	Not enough inference steps	Bump Inference steps to 40+
Model ignores your prompt keywords	Guidance too low	Bump Guidance scale to 8–10
All takes sound the same	Seed pinned + same params	Turn Random seed on; increase Batch size
First generation takes 90s	Model warming up in memory	Normal — subsequent runs are much faster
'Backend offline' badge	Server crashed or port collision	Ctrl-C in terminal → <code>ls -ti :8001 :5173 xargs kill -9</code> → <code>./start.sh</code>
'LLM init failed' toast on Enhance	LM not loaded	Add <code>ACESTEP_INIT_LLM=true</code> to your .env, restart
Generation succeeded but no audio	Empty output from model (rare)	Re-run — usually a bad seed
Out-of-memory on Apple Silicon	Duration too long or batch too big	Lower Duration to 60s, Batch size to 1, keep Tiled decode on

8. What each sidebar page is for

Page	What it does
Studio / Generate	Where you make music (same page).
Projects	Your generations grouped by title. Rename projects by setting a Project title before generating.
History	Chronological list of every run in this browser. Play, download, delete, re-open.
Audio Library	Every successful track from every project, searchable.
Settings	Backend URL, optional API key, default format, default steps / guidance / duration, clear local data.
About	Version, credits, and honest notes on limitations.

Note: history is local to your browser (in localStorage). Clearing site data wipes it. The audio files themselves live on the backend and aren't affected.

9. Progression path — beginner to power user

Day 1

Custom mode, click Random sample, generate. Play with prompts.

Day 2 to 3

Bring your own lyrics. Tag them. Try Enhance caption. Compare batches.

Week 1

Learn what BPM / key / steps / guidance actually change by A/B testing (same prompt, one field different).

Week 2

Try Remix on a demo you recorded. Try Repaint to fix a specific verse.

Later

Turn on Think mode. Play with SDE inference. Try DCW wavelets in Advanced. Start using seeds to reproduce takes.

End of guide.

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